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ARC 101: Introduction to Architecture - Design Thinking, Drawing and Site

From the design process to scale drawings, orientation and the language of buildings

FACULTY PRIMER

Faculty Primer

Faculty of Environmental Sciences
Architecture | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

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Why this pack exists

Architecture freshers meet two demands at once: thinking conceptually and drawing precisely. This pack explains the design process, how to read and produce scale drawings, site and climate considerations, and the vocabulary you need to critique a building intelligently.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe the stages of the architectural design process from brief to detail.
- Read and produce measured drawings using scale, plans, sections and elevations.
- Analyse a site for orientation, drainage, access, noise and climate.
- Use architectural vocabulary accurately when presenting or critiquing work.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

The design process

Design is not decoration applied at the end. It is a sequence of decisions from problem to built response.

- Brief: what the client needs, who will use the building, what constraints exist.
- Site analysis: orientation, access, topography, vegetation, noise, views and services.
- Concept: the organising idea - a diagram, a spatial strategy, a response to climate or culture.
- Schematic design: plans, sections and massing at small scale.
- Design development: materials, structure, services and detail coordination.
- Presentation: drawings, physical or digital models, and a clear verbal explanation.

Concept is not decoration
A concept explains why the building is organised the way it is. 'I used curves because they are beautiful' is a preference, not a concept.

Drawing and scale

Architectural drawing is a precise language. Learn the conventions and your work becomes legible to every reviewer.

- Plan: a horizontal cut about 1.2 metres above floor level.
- Section: a vertical cut showing heights and construction.
- Elevation: an external face of the building, without perspective.
- Scale: 1:100 or 1:50 for design drawings; 1:20 or 1:5 for details.
- Line weights: thin for construction lines, heavy for cut elements, medium for visible edges.
- Always include a north arrow, scale bar, title block and human figure for scale.

Site, climate and comfort

In a hot-humid climate such as Awka, passive design is not optional. Orientation and shading decide whether a room is usable.

- Long axis east-west reduces direct sun on long facades.
- Deep verandahs, brise-soleil and shaded openings reduce heat gain.
- Cross ventilation needs inlet and outlet openings on opposite sides.
- Consider prevailing winds, rainfall intensity and drainage before placing openings.
- Access and circulation: separate pedestrian and vehicle entry, and respect setbacks.
- Check the site's slope; it determines plinth levels and drainage strategy.

Critique language
Say 'the circulation is legible' or 'the service core is poorly located'. Avoid 'nice' and 'ugly'. Describe the effect, then judge it.

Architecture as response to conditions

A building is an answer to a set of conditions: climate, site, materials, budget, function, culture and regulation. Architectural quality is not decoration applied afterwards; it is how well the answer addresses those conditions at once.

In Nigeria the conditions are specific and demanding. Heat and humidity require shading, cross-ventilation and thermal mass; heavy rain requires overhangs and drainage; unreliable power requires daylighting strategies; and informal settlement requires incremental, affordable construction rather than imported models.